



Ansys Fluent Simulation Report

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System Information

Application	Fluent
Settings	2d, density-based implicit, inviscid
Version	25.2.0-10204
Source Revision	5eecd5d865
Build Time	Jun 16 2025 10:48:28 EDT
CPU	11th Gen Intel(R) Core(TM) i5-11400H @ 2.70GHz
OS	Windows

Geometry and Mesh

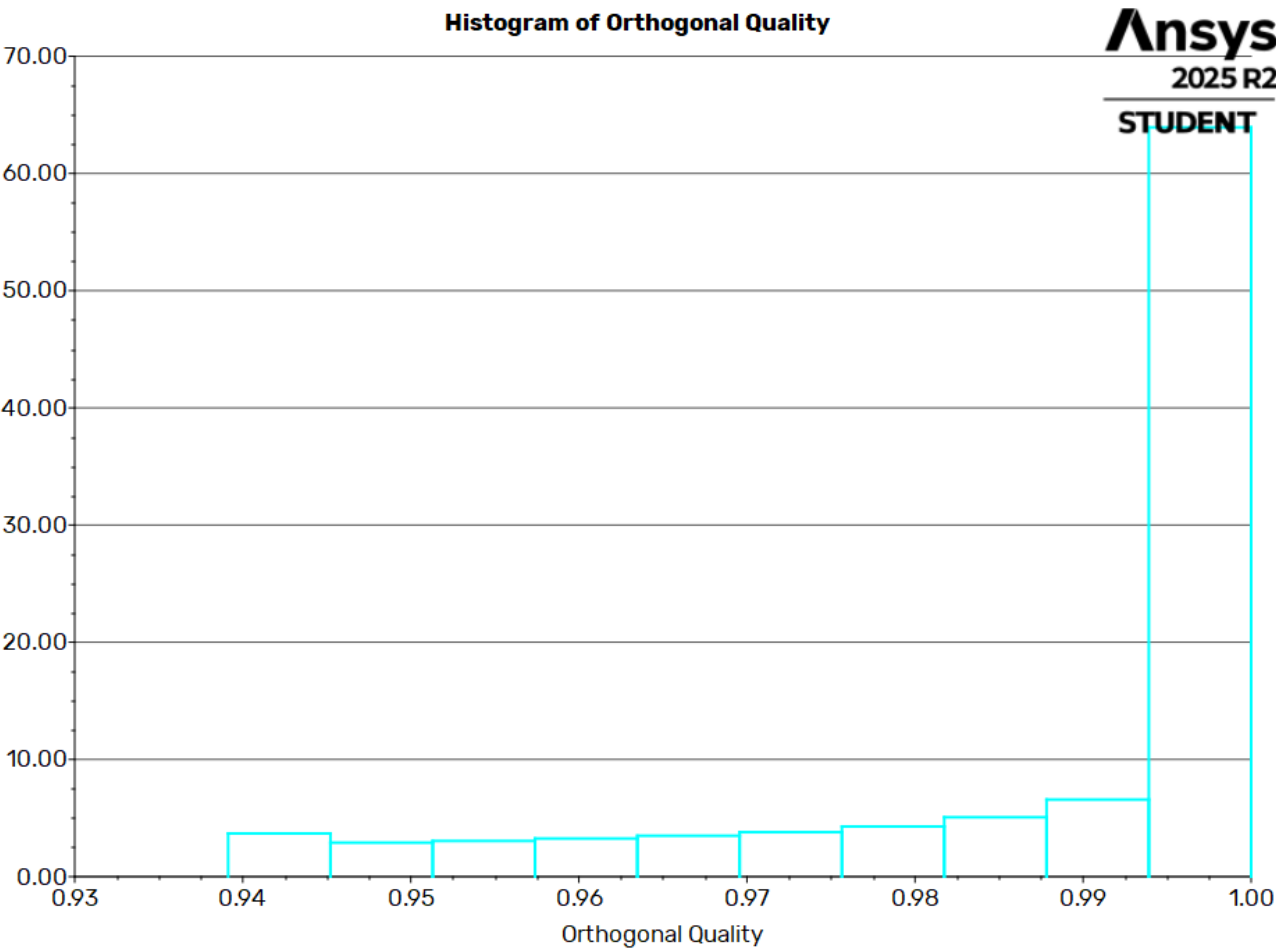
Mesh Size

Cells	Faces	Nodes
13837	27912	14076

Mesh Quality

Name	Type	Min Orthogonal Quality	Max Aspect Ratio
compressioncorner	Quad Cell	0.93913025	2.119554

Orthogonal Quality



Simulation Setup

Physics

Models

Model	Settings
Space	2D
Time	Steady
Viscous	Inviscid
Heat Transfer	Enabled

Material Properties

– Fluid	
– air	
Density	ideal gas
Cp (Specific Heat)	1006.43 J/(kg K)
Molecular Weight	28.966 kg/kmol
– Solid	
aluminum	

Cell Zone Conditions

– Fluid	
– compressioncorner	
Material Name	air
Specify source terms?	no
Specify fixed values?	no
Frame Motion?	no
Porous zone?	no

Boundary Conditions

– Inlet	
– pressure_outlet	
Gauge Pressure [Pa]	101325
Mach Number	3
Temperature [K]	288
Component of Flow Direction (x,y)	(1, 0)
– pressure_inlet	
Gauge Pressure [Pa]	101325
Mach Number	3
Temperature [K]	288
Component of Flow Direction (x,y)	(1, 0)
– Wall	
boundary	wall
wall	wall

Reference Values

Area	1 m ²
Density	1.225 kg/m ³
Depth	1 m
Enthalpy	0 J/kg
Length	1 m
Pressure	0 Pa
Temperature	288.16 K
Velocity	1 m/s
Ratio of Specific Heats	1.4
Yplus for Heat Tran. Coef.	300
Reference Zone	compressioncorner

Solver Settings

– Equations	
Flow	True
– Numerics	

Absolute Velocity Formulation	True
Under-Relaxation Factors	
– Discretization Scheme	
Flow	Second Order Upwind
– Time Marching	
Solver	Implicit
Courant Number	5
– Solution Limits	
Minimum Absolute Pressure [Pa]	1
Maximum Absolute Pressure [Pa]	5e+10
Minimum Static Temperature [K]	1
Maximum Static Temperature [K]	5000

Run Information

Number of Machines	1
Number of Cores	2
Case Read	6.534 seconds
Iteration	1.894 seconds
AMG	0.761001 seconds
Virtual Current Memory	0.974953 GB
Virtual Peak Memory	1.00126 GB
Memory Per M Cell	24.4238

Solution Status

Iterations: 134

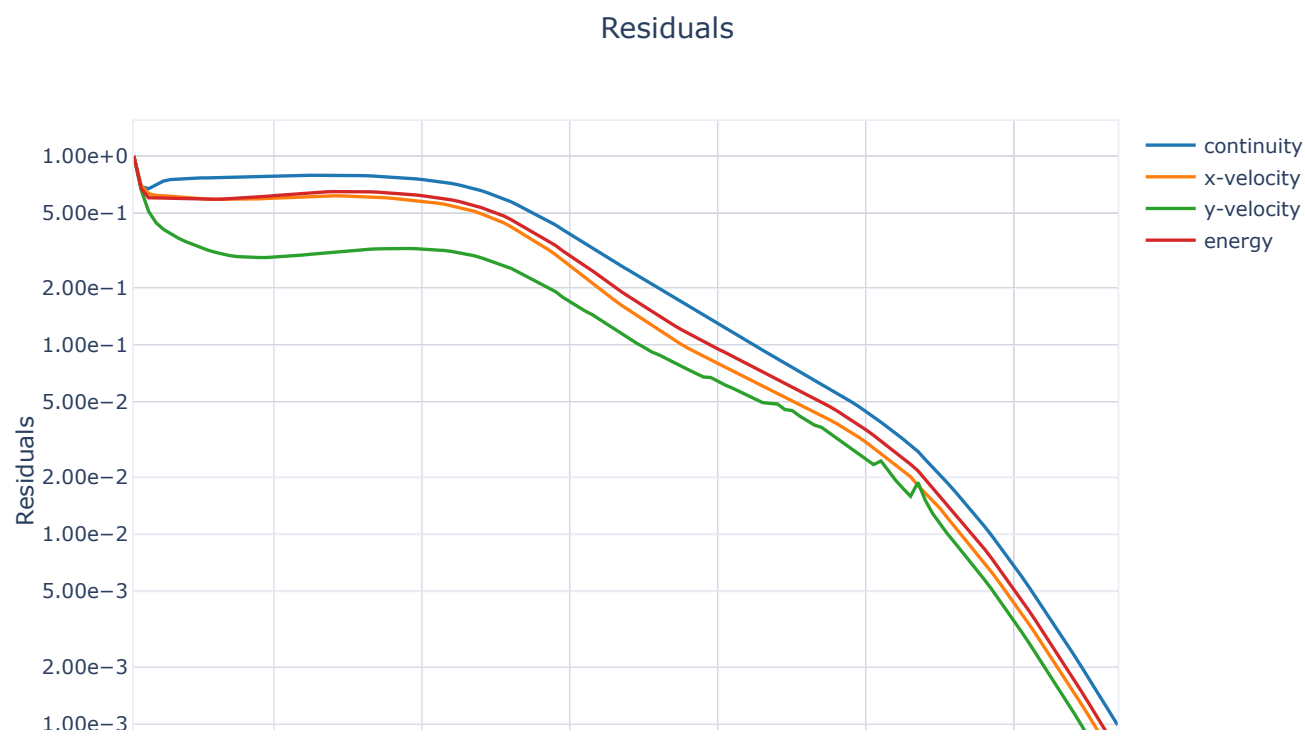
	Value	Absolute Criteria	Convergence Status
continuity	0.000983051	0.001	Converged
x-velocity	0.0006190325	0.001	Converged
y-velocity	0.000467324	0.001	Converged
energy	0.0007206539	0.001	Converged

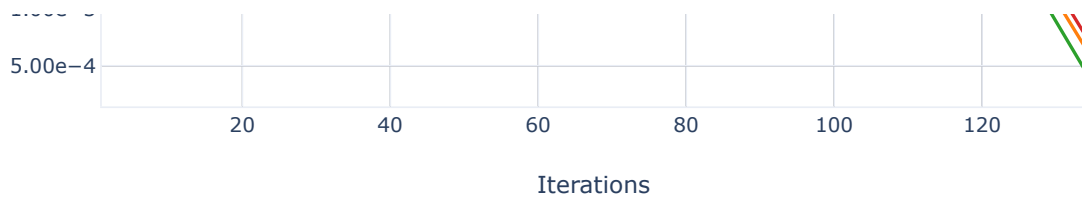
Report Definitions

cfl-number	5
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Plots

Residuals

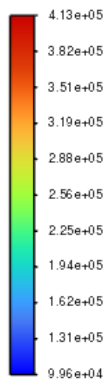




Contours

pressure_contours

Static Pressure
[Pa]



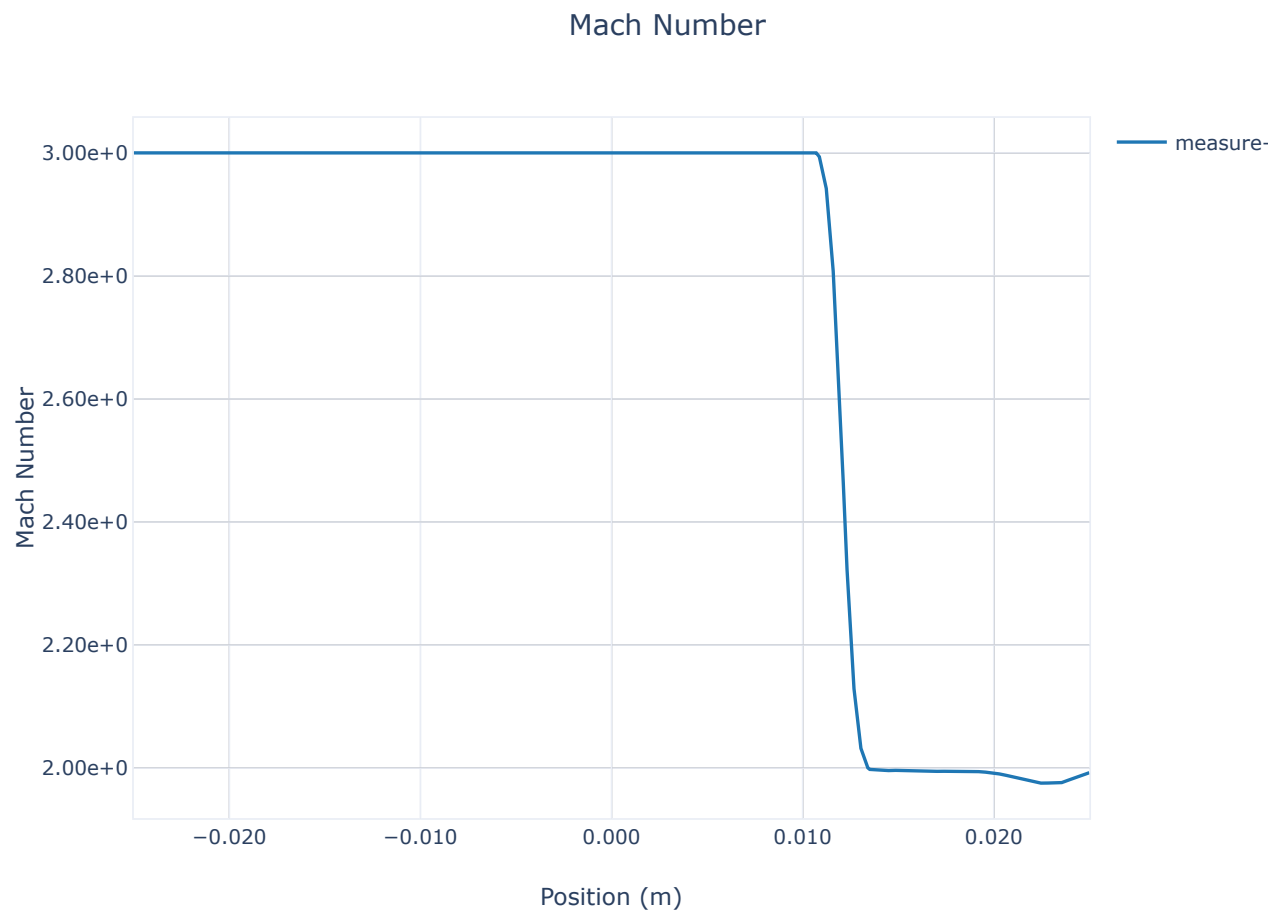
pressure_contours



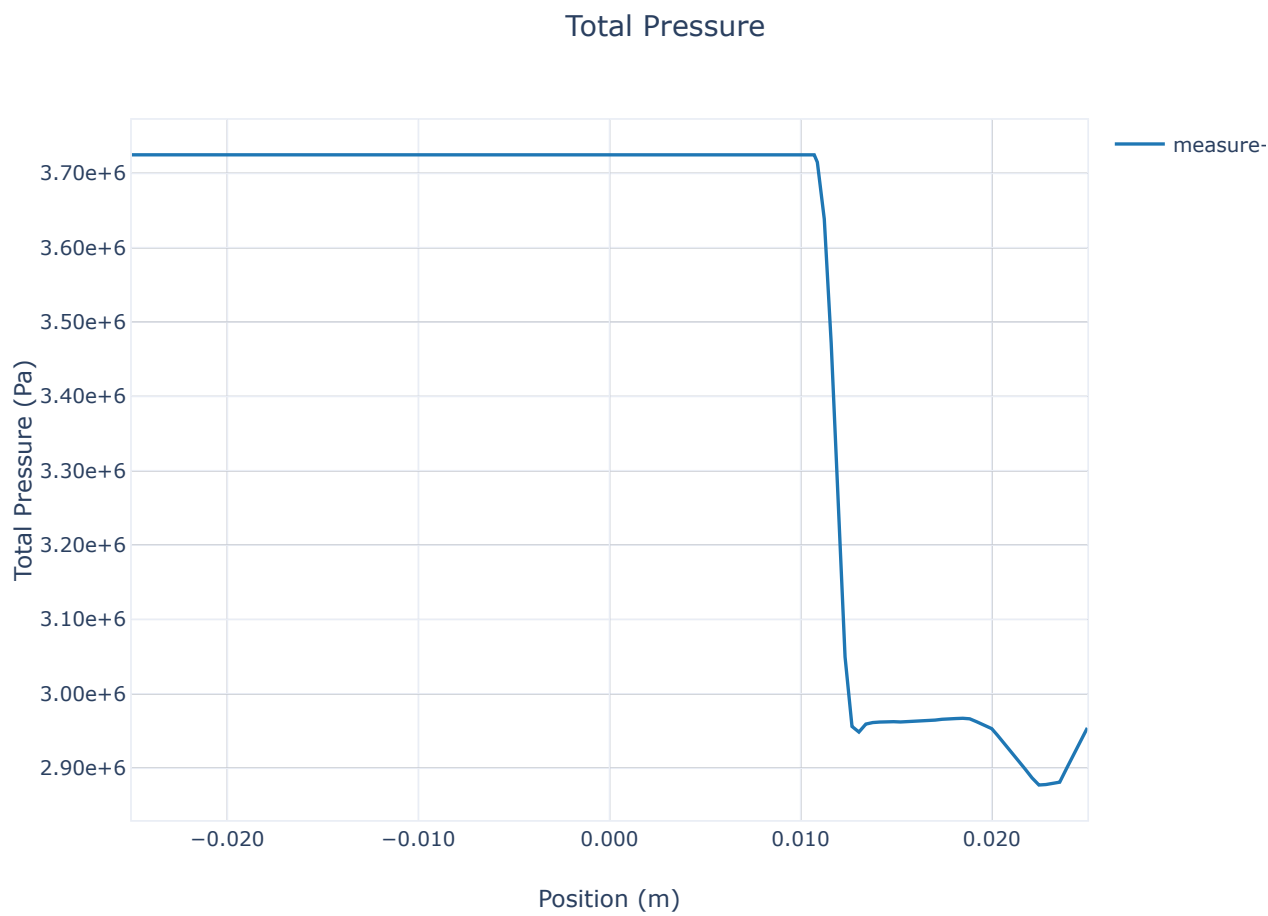
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XY Plots

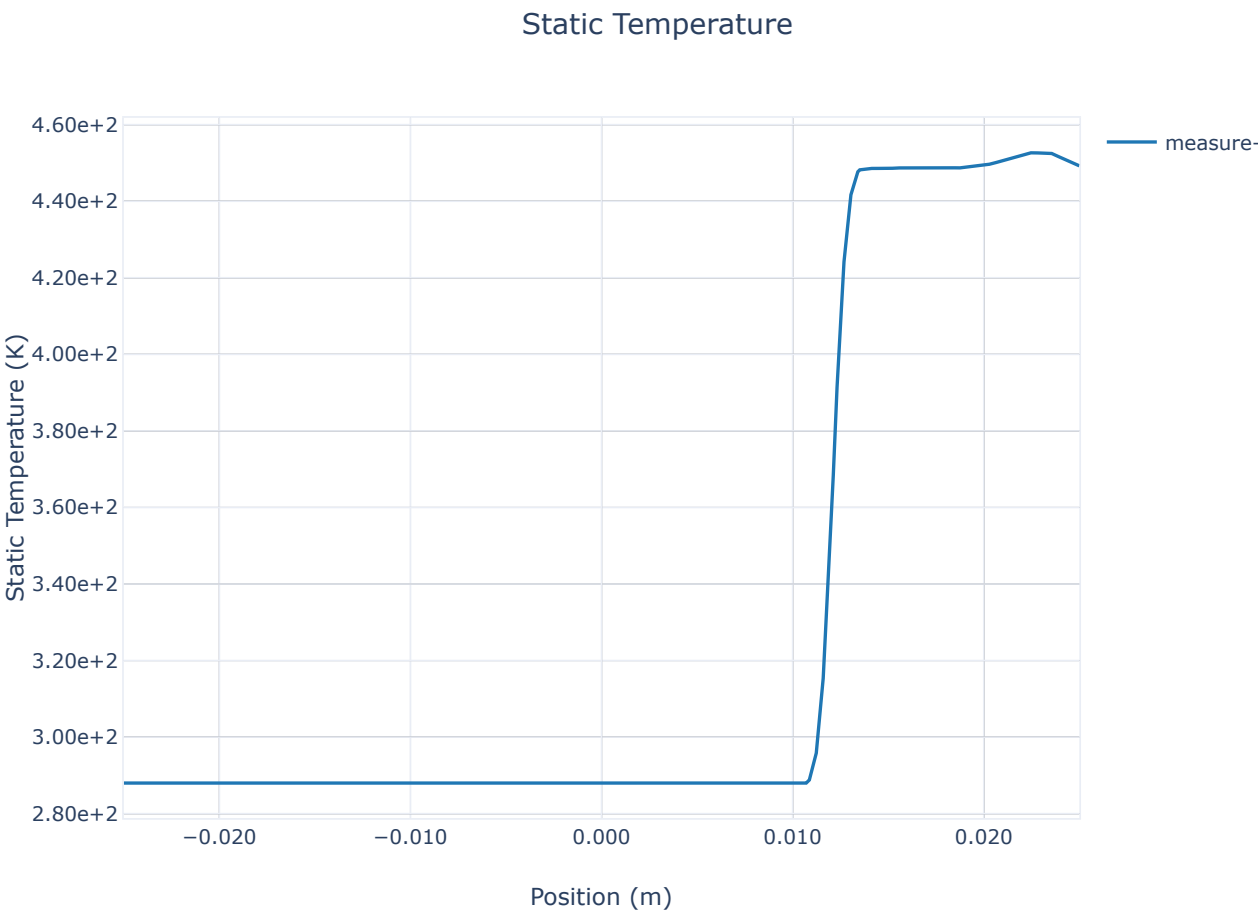
mach-no



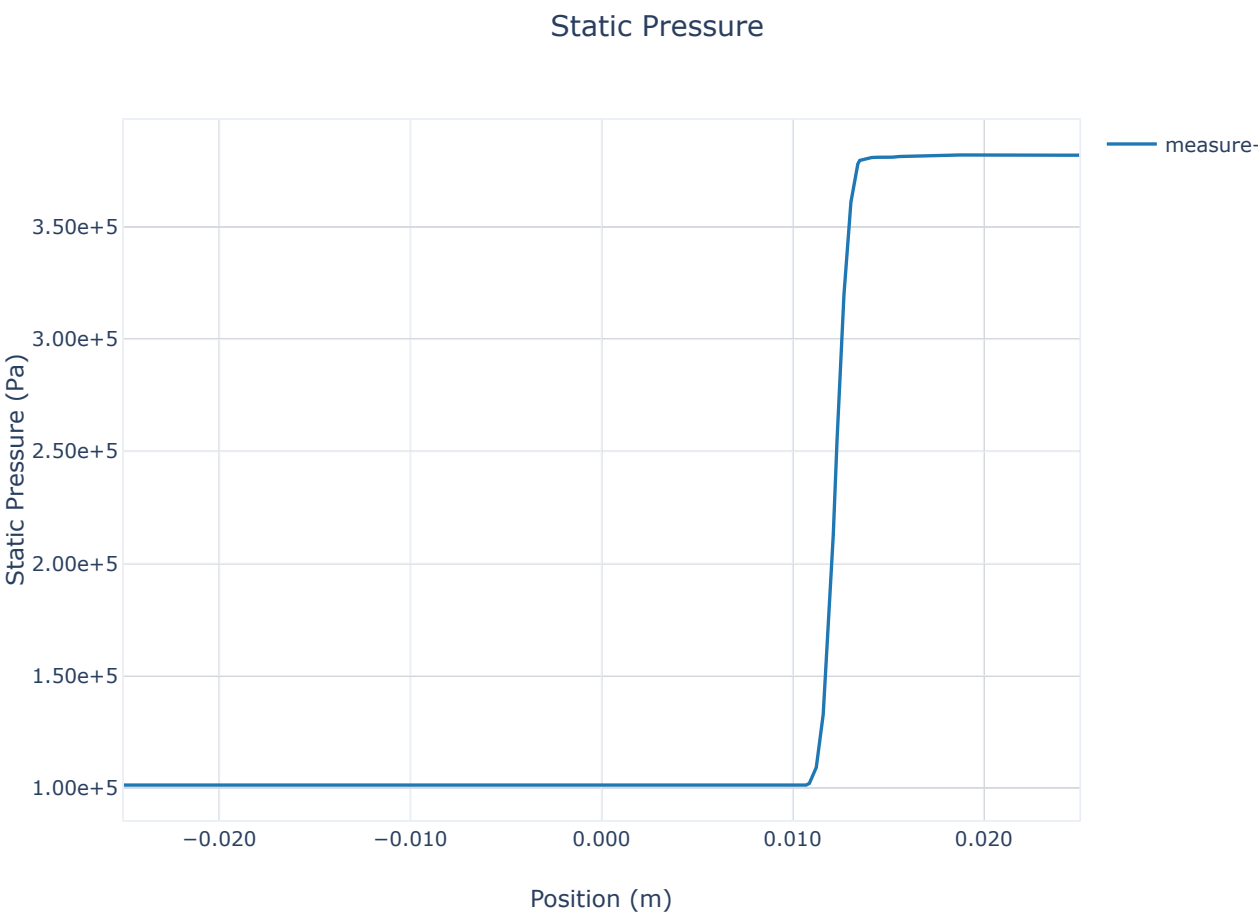
total-pressure



tempreature-plot

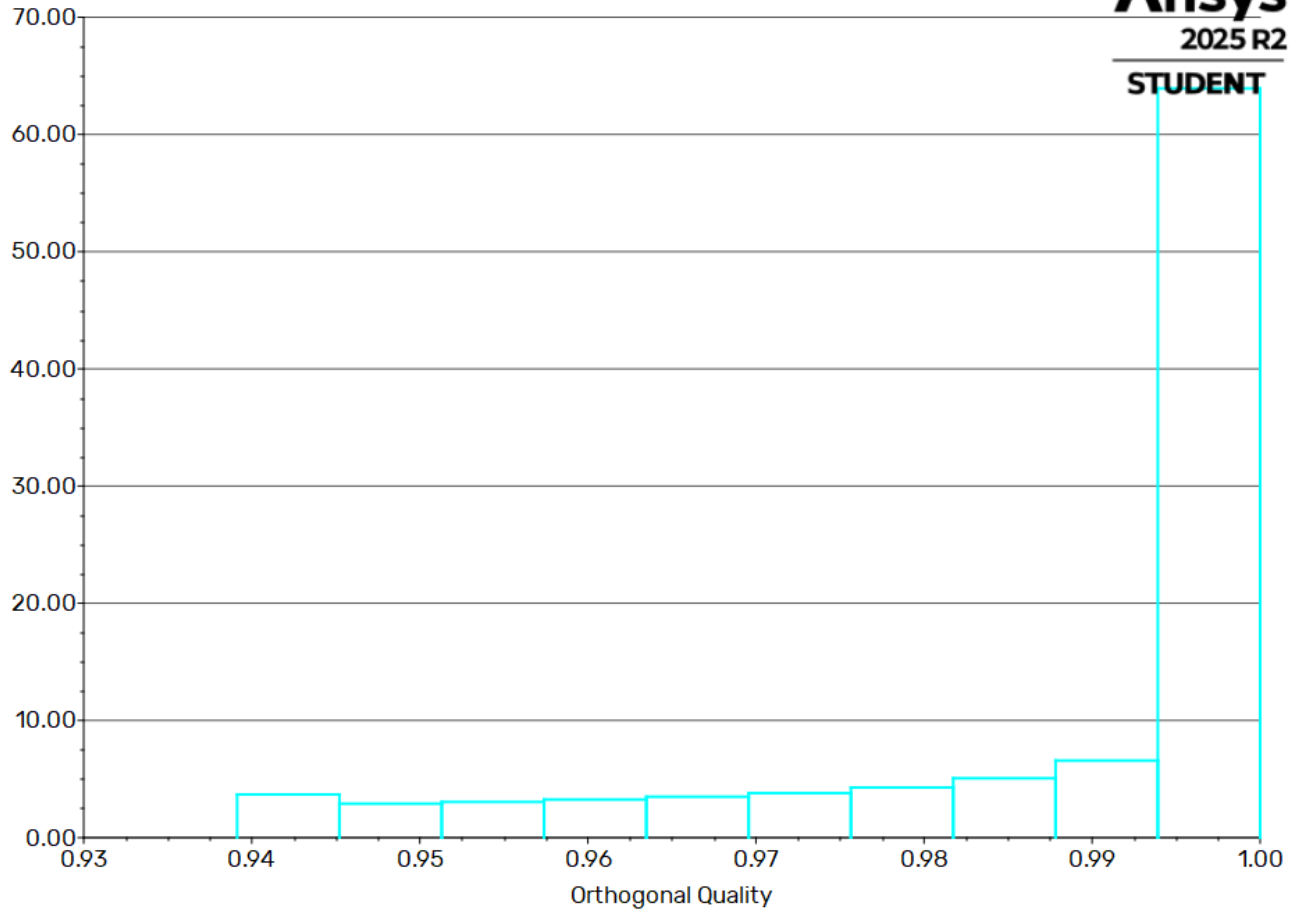


pressure_plot

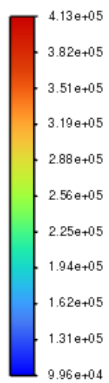


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Histogram of Orthogonal Quality



Static Pressure
[Pa]



pressure_color

